

Homework 2: August 2025

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You do not have to submit this assignment.

Exercise 2.1. A randomized algorithm¹ is one for which the output of the algorithm is a random variable. Suppose that we have a randomized algorithm for a problem that outputs the correct answer with probability $0.5 + \delta$ (for some $\delta > 0$). Assume that if we run the algorithm on the same input multiple times, the corresponding outputs are independent and in every run, the output is correct with probability $0.5 + \delta$.

Is it possible to achieve an accuracy (probability of correct answer) to $1 - \epsilon$ for any arbitrary $\epsilon > 0$ by repeating the algorithm multiple times and somehow processing the outputs? How many times would you need to repeat this?

Exercise 2.2. Verify whether the following distributions are subgaussian and/or subexponential:

1. Bernoulli(p)
2. Binomial(n, p)
3. Poisson(α)
4. Gaussian(μ, σ^2)
5. Gamma(k, θ)
6. Exponential(α)

For the following exercise, you may use the Cauchy-Schwartz inequality without proof:

$$|\mathbb{E}[XY]| \leq (\mathbb{E}X^2)^{1/2} (\mathbb{E}Y^2)^{1/2}$$

Exercise 2.3. Let $X_1, X_2, \dots$ be an infinite sequence of random variables (not necessarily independent), where each X_i is subgaussian with subgaussian norm ν_i . Prove that

$$\mathbb{E} \left[\max_i \frac{|X_i|}{\sqrt{1 + \log i}} \right] \leq c \max_i \nu_i,$$

where c is a universal constant. Now, assume that $\nu_i = \nu$ for all i and that the random variables are independent. Prove that

$$X = \sum_{i=1}^{\infty} \frac{X_i}{2^i}$$

is subgaussian with subgaussian norm ν .

For general (possibly dependent) X_1, X_2 that are subgaussian with subgaussian norms ν_1, ν_2 respectively, show that $X_1 + X_2$ is subgaussian with subgaussian norm at most $\sqrt{2(\nu_1^2 + \nu_2^2)}$.

¹https://en.wikipedia.org/wiki/Randomized_algorithm

2.0.0.1 Isotropic random vectors

A random vector $\underline{X}$ in $\mathbb{R}^d$ is said to be isotropic if

$$\mathbb{E}[\underline{X}\underline{X}^T] = I_d,$$

where I_d denotes the $d \times d$ identity matrix.

Exercise 2.4. • Give an example of a random vector with non-i.i.d. components in $\mathbb{R}^2$ that is isotropic.

- Show that two $d \times d$ symmetric matrices A, B are equal iff $\underline{x}^T A \underline{x} = \underline{x}^T B \underline{x}$ for every vector $\underline{x} \in \mathbb{R}^d$. Is this statement true if A, B are not symmetric?
- Show that $\underline{X}$ is isotropic if and only if $\mathbb{E}[\langle \underline{x}, \underline{X} \rangle^2] = \|\underline{x}\|_2^2$ for all (deterministic) $\underline{x} \in \mathbb{R}^d$.
- Show that if $\underline{X}$ has jointly Gaussian components and is isotropic, then the components are independent standard normal.
- If $\underline{X}$ is isotropic, what is $\mathbb{E}\|\underline{X}\|_2^2$? If $\underline{X}, \underline{Y}$ are independent and isotropic, then what is $\mathbb{E}[\langle \underline{X}, \underline{Y} \rangle^2]$?
- Let $\underline{X}$ be isotropic Gaussian, with zero mean and unit variance components. Represent $\underline{X}$ in polar form as $\underline{X} = r\theta$, where $r = \|\underline{X}\|_2$, and θ is a unit norm vector. What are the distributions of r and θ ? Are r, θ independent?

2.0.0.2 Configuration functions

This section makes use of the Efron-Stein inequality (see Sec 3.1 in “Concentration Inequalities” by Boucheron, Lugosi and Massart).

Let $\mathcal{X}$ be an arbitrary set, and Π be a property defined on finite products of $\mathcal{X}$, i.e., Π can be defined as a collection of sets $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3, \dots$, where $\mathcal{S}_k \subset \mathcal{X}^k$ for $k = 1, 2, 3, \dots$. We say that a sequence/vector $\underline{x} \in \mathcal{X}^m$ satisfies property Π if $\underline{x} \in \mathcal{S}_m$.

A property Π is said to be hereditary if it satisfies the following property: if $\underline{x} = (x_1, \dots, x_m)$ satisfies property Π , then every subsequence of $\underline{x}$ also satisfies Π , or equivalently,

$$(\underline{x} \in \mathcal{S}_m) \implies (\underline{x}_{i_1, \dots, i_k} \in \mathcal{S}_k) \text{ for all } 1 \leq k \leq m \text{ and } i_1 < i_2 < \dots < i_k$$

The configuration function associated with such a Π , denoted f_Π , is the function that maps every $\underline{x}$ to the size of the largest subsequence of $\underline{x}$ that satisfies Π .

For a simple example, assume that $\mathcal{X} = \mathbb{Z}$, and $\mathcal{S}_k = 2\mathbb{Z}^k$ for $k = 1, 2, \dots$. In this case, $\underline{x}$ satisfies property Π iff $\underline{x}$ is a sequence of even integers. The configuration function $f_\Pi(\underline{x})$ outputs the size of the largest subsequence of $\underline{x}$ consisting only of even numbers, or equivalently, the number of even integers in $\underline{x}$. E.g., $f_\Pi(1, 2, 4, -5, 0, 98) = 4$.

Exercise 2.5. Prove that if f is a configuration function and $X_1, \dots, X_n$ are independent random variables, then

$$\text{Var}(f(X_1, \dots, X_n)) \leq \mathbb{E}[f(X_1, \dots, X_n)]$$

Exercise 2.6. For any sequence $x_1, \dots, x_n$, let $f(x_1, \dots, x_n)$ denote the number of distinct elements in $x_1, \dots, x_n$. For example, $f(01101010) = 2$ and $f(10026001) = 4$.

Let p_X be an arbitrary pmf over the set of positive integers, and $X_1, \dots, X_n$ be iid random variables with pmf p_X .

- Does f satisfy the bounded differences property? If so, find $c_1, \dots, c_n$. Use this to get an upper bound for the variance of $Z = f(X_1, \dots, X_n)$.
- Compute an expression for $\mathbb{E}f(X_1, \dots, X_n)$ in terms of p_X . How does this scale with n ? Linearly? Sublinearly? If you try to use your bound for the variance from the first part and apply Chebyshev's inequality, what bound can you derive for

$$\Pr[|Z - \mathbb{E}[Z]| \geq \delta \mathbb{E}[Z]]?$$

- Prove that f is a configuration function. Using the previous exercise and Chebyshev's inequality, improve the tail bound in the previous part. For what values of δ does this give a meaningful bound?